

Executive Summary Report: E-commerce & Netflix Titles Data Analysis

Analysis Objective: To clean, explore, and extract actionable business insights from two distinct datasets (E-commerce retail transactions and the Netflix content catalog) to understand customer purchasing behaviors and content distribution trends.

1.1 Key Metrics

1.1.1 E-commerce Dataset

These are key metric findings for revenue, items sold, and customer purchase pattern;

Metric	Value
Total Revenue	£ 8,887,208.89
Total items sold	5,152,002
Active Unique Customer	4,338

International Market Sales Analysis (Rank)

Based on the analysis, Netherlands market sales trends depicts highest sales performance.

Market Trends	Rank
Netherlands	1
EIRE	2
Germany	3
France	4

Australia	5
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1.1.2 Netflix Titles

Top Content Producing Countries by Rank

Country	Rank
USA	1
India	2
Uk	3
Japan	4
South Korea	5

1.2 Data Cleaning Challenges & Resolutions

Before analysis, both datasets required rigorous preprocessing to ensure data integrity. The most significant challenges included:

Dataset	Challenges	Resolutions
E-commerce (Online Retail)	135,050 transactions were missing a CustomerID in the E-commerce dataset	These rows were dropped, as customer identity is mathematically required for behavioral and retention analysis
E-commerce (Online Retail)	The dataset contained over 10,000 rows with negative Quantity values (representing canceled orders/returns) or zero pricing	The negative values were filtered out to prevent data bias and negative revenue calculations
Netflix_titles	The Netflix dataset also have from heavy missing data in the director, cast, and country	The missing data was filled with "Unknown ' to preserve thedataset's overall shape, nstead of dropping it

	columns (over 4,000 instances).	
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1.3 Key Exploratory Data Analysis (EDA) Findings

1.3.1 E-commerce Purchasing Patterns and Netflix Content Analysis:

Dataset	Exploratory Data Analysis (EDA) Findings
E-commerce (Online Retail)	Revenue exhibits a massive spike in November, indicating a business model heavily reliant on Q4 holiday/end-of-year shopping.
E-commerce (Online Retail)	High-volume items like "WORLD WAR 2 GLIDERS" and "JUMBO BAG RED RETROSPOT" lead in quantity, but the vast majority of catalog items are low-ticket goods priced under £3.
E-commerce (Online Retail)	A select few "VIP" customers (e.g., ID 14646) generate a disproportionately high percentage of total revenue, suggesting strong B2B or wholesale relationships
Netflix Titles	The platform heavily favors feature-length content, with Movies making up roughly 70% of the catalog compared to TV Shows (30%)
Netflix Titles	The most common content rating by a wide margin is TV-MA (Mature Audiences), signaling a strategic shift away from family-centric programming toward adult-oriented media.
Netflix Titles	While the United States is the top content producer, India holds a remarkably strong second place, dominating the "International Movies" genre

1.4 Top Actionable Insights and Recommendations

Insights	Recommendation
Capitalize on E-commerce seasonality	Inventory for top-performing items (like novelty gliders and bags) must be aggressively stocked by late Q3 to prepare for the November revenue surge. Marketing budgets should be heavily weighted toward this window.
Few customers drive massive revenue	The business should assign dedicated account managers or implement tiered bulk-discount pricing to retain these high-value buyers.
India positioned has top-two content producer especially in dramas and comedy	Netflix should continue investment in localized Indian content (especially in the Drama and Comedy genres) which represents a massive avenue for international subscriber retention and growth
The data shows Netflix's catalog is fundamentally adult-driven.	Production budgets should continue to prioritize TV-MA International Movies and Dramas, as this is where the core catalog strength lies.